

Program and Process

21. A process in OS is:
- a) Program in execution
 - b) Data file
 - c) Register
 - d) Interrupt signal
22. Program code is stored in:
- a) .data section
 - b) .text section
 - c) .heap section
 - d) .bss section
23. The difference between program and process:
- a) Program is passive, process is active
 - b) No difference
 - c) Both are stored in memory
 - d) Process is passive, program is active
24. Stack is used to store:
- a) Data only
 - b) Instructions only
 - c) Function calls and local variables
 - d) Only binaries
25. Heap section is used for:
- a) Static allocation
 - b) Dynamic memory allocation
 - c) Code storage
 - d) Register values

Sections

26. Which section contains code instructions?
- a) .data
 - b) .bss
 - c) .text
 - d) .stack
27. Uninitialized global variables are placed in:
- a) .data
 - b) .bss
 - c) .text
 - d) .code
28. Stack frame contains:
- a) Return address
 - b) Local variables
 - c) Function arguments

d) All of the above

29. Initialized data is stored in:

- a) .bss
- b) .text
- c) .data
- d) .stack

30. Local variables created inside functions reside in:

- a) Heap
- b) Stack
- c) .data
- d) .text

Address Space

31. Kernel address space is used by:

- a) User programs
- b) Operating system
- c) Peripheral devices
- d) Hard disk

32. User address space holds:

- a) Kernel code
- b) User applications
- c) BIOS
- d) Device drivers

33. Privilege separation is important for:

- a) Performance
- b) Security
- c) Cost
- d) Size

34. Access to kernel space requires:

- a) Privileged instructions
- b) User mode
- c) RAM access
- d) Magnetic storage

35. Context switch involves switching between:

- a) Devices
- b) Processes
- c) Registers
- d) Instruction sets

Booting

36. BIOS is responsible for:

- a) Program execution
- b) Power on self test

- c) Data transfer
 - d) Print preview
37. Bootloader loads:
- a) Application
 - b) OS kernel
 - c) User profile
 - d) User data
38. Which is loaded last in the boot sequence?
- a) BIOS
 - b) Bootloader
 - c) Operating System
 - d) Applications
39. BIOS stands for:
- a) Basic Input Output System
 - b) Binary Input Output System
 - c) Basic Integrated Operation System
 - d) Boot Integrated Output System
40. The main function of OS during boot is to:
- a) Initialize peripherals
 - b) Load kernel
 - c) Run user applications
 - d) Format hard disk

Context Switch and Context

41. Context involves saving/restoring:
- a) GPR
 - b) Flags
 - c) PC (Program Counter)
 - d) All of the above
42. Stack pointer holds address of:
- a) Stack base
 - b) Next instruction
 - c) Top of stack
 - d) Code section
43. Flags register stores:
- a) Status bits
 - b) Addresses
 - c) Data values
 - d) Function calls
44. Context switching is essential for:
- a) Multitasking
 - b) Single tasking
 - c) RAM management

d) Code optimization

45. Privileged instructions are executed in:

- a) User mode
- b) Kernel mode
- c) Application mode
- d) Stack frame

Stack, Privilege, Toolchain

46. Stack frame is created when:

- a) Program starts
- b) Function is called
- c) Stack is cleared
- d) Memory is freed

47. Toolchain typically includes:

- a) Compiler, Debugger, Linker
- b) Compiler only
- c) Linker only
- d) Hardware only

48. System calls provide interface to:

- a) Kernel
- b) User applications
- c) BIOS
- d) RAM

49. fopen in C is a:

- a) Library call
- b) System call
- c) Context switch
- d) Interrupt

50. objdump is used to:

- a) Build code
- b) Disassemble binaries
- c) Debug Java
- d) Format disk